CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. Examination (CE/IT)

November – December 2011

CE 204 Digital Electronics

Date: 01.12.2011, Thursday **Time: 01:30 p.m. To 04:30 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) 10111 is a sign binary number, determine decimal value using (i) sign magnitude (ii) 1's complement (ii) 2's complement method. **[03]**

(b) "Each minterm can be an implicant but each implicant need not be a minterm" Justify. **[04]**

Q - 2 (a) Find out 5's complement of $(455)_6$ **[02]**

(b) Simplify the function given below using tabulation method: **[06]**

$$f(A,B,C,D,E,F,G)=\sum(20,28,52,60)$$

(c) State DeMorgan's theorems and demonstrate by means of truth tables validity of it. **[06]**

OR

Q - 2 (a) Show that the dual of the exclusive-OR is equal to its complement. **[02]**

(b) Simplify the Boolean function $f(w,x,y,z)=\sum(0,1,2,4,5,6,8,9,12,13,14)$ using K-map.Realize the simplified function using only NAND gates, if both normal and complement inputs are available. **[06]**

(c) Express the following functions in sum of minterms and product of maxterms. **[06]**

(1) $F(x,y,z) = (xy + z)(y+xz)$

(2) $F(x,y,z) = x + y$

Q - 3 (a) Design half subtractor using decoder and EX-OR. **[02]**

(b) What do you mean by universal gate? Prove that NOR as a universal gate. **[04]**

(c) What is difference between ROM and PLA? **[08]**
$F_1=A'B+AB'$, $F_2=A'B'+AB$ Implement using PLA and make a program table.

OR

Q - 3 (a) Using 2's complement method subtract $(1011)_2 - (0100)_2$ **[02]**

(b) Implement following function $f(A,B,C,D)=\sum(0,1,3,4,8,9,15)$ using multiplexer. **[04]**

(c) A company has four directors A,B,C and D.The percentage of shares held by them are 35%, 30%, 25% and 10% respectively. Their voting power is proportional to shares held by them. Any major decision must have a support of 60% or above of the stock. Design a combinational logic circuit for voting in the company. **[08]**

SECTION – II

Q - 4 (a) What is race around condition? How can we remove it? [03]

(b) Give the Difference between a decoder and demultiplexer. How a decoder can be used as a demultiplxer? [04]

Q - 5 (a) Design sequential circuit from state table shown in Figure 1.(Use T-flipflop) [07]

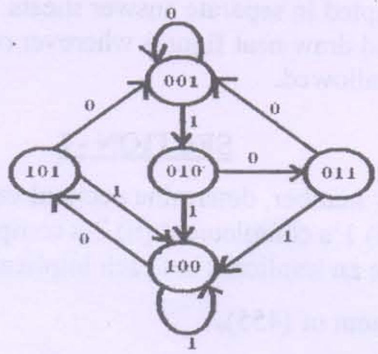

Figure 1. State diagram

(b) With neat sketch explain bidirectional shift register. [07]

OR

Q - 5 (a) Design synchronous BCD counter. [07]

(b) Discuss D-type edge-triggered flip-flop in detail. [07]

Q - 6 (a) (i) Give the difference between synchronous counter and Asynchronous counter [03]

(ii) Define following terms with respect to sequential circuit [04]

(1) State (2) State table (3) Characteristic table (3) Excitation table

(b) Design mod-6 counter with following binary repeated sequence:0,1,2,4,5,6.·(Use T flip flop). [07]

OR

Q - 6 (a) Explain in brief ring counter and Johnson counter. [04]

(b) (i) What is disadvantage of RS flip-flop? [01]

(ii) Make characteristic and excitation table of RS,JK,D and T flip-flop. [04]

(iii) With neat sketch explain the operation of clocked JK flip-flop. [05]

****Best Luck****